Introduction to Statistical Methods

(S2-25_AIMLCLZC418) – Assignment 1

AIML – Section 4

Question 1&2 carries 2.5 marks each.
Question 3 carries 5 marks
Total - 10 Marks
Duration: 30th May 2026 – 14th June 2026
1) Submissions are individual
2) Solve these on paper, scan, and upload
3) Plagiarism results in zero marks
4) Write your name, BITS ID and Section on each page
5) Only handwritten solutions with formula, full steps with proper justification are required.

Answer all the questions

1. Given

$$f(x) = ax^2, 0 < x < 1$$
$$= 0, elsewhere$$

Find the constant a. Also find distribution function $F(x)$, mean and variance of X. **[2.5 M]**

2. Find the probability that out of 100 patients between 84 and 95 inclusive will survive a heart-operation given that the chances of survival is 0.9. **[2.5 M]**

3. A lighting company stores light bulbs in three different boxes for quality testing.

   Box I contains 10 bulbs, out of which 4 are defective.

   Box II contains 6 bulbs, out of which 1 is defective.

   Box III contains 8 bulbs, out of which 3 are defective.

A quality inspector randomly selects one box and then randomly picks one bulb from that box. The selected bulb is found to be defective.

Based on this information, answer the following questions.

(a) Find the probability of selecting:

1. Box I
2. Box II

3. Box III

(b) Find the probability that the selected bulb is defective given that Box I was selected.

(c) Find the probability that the selected bulb is defective given that Box II was selected.

(d) Find the probability that the selected bulb is defective given that Box III was selected.

(e) If the selected bulb is defective, find the probability that it came from:

1. Box I
2. Box II
3. Box III **[5 M]**

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